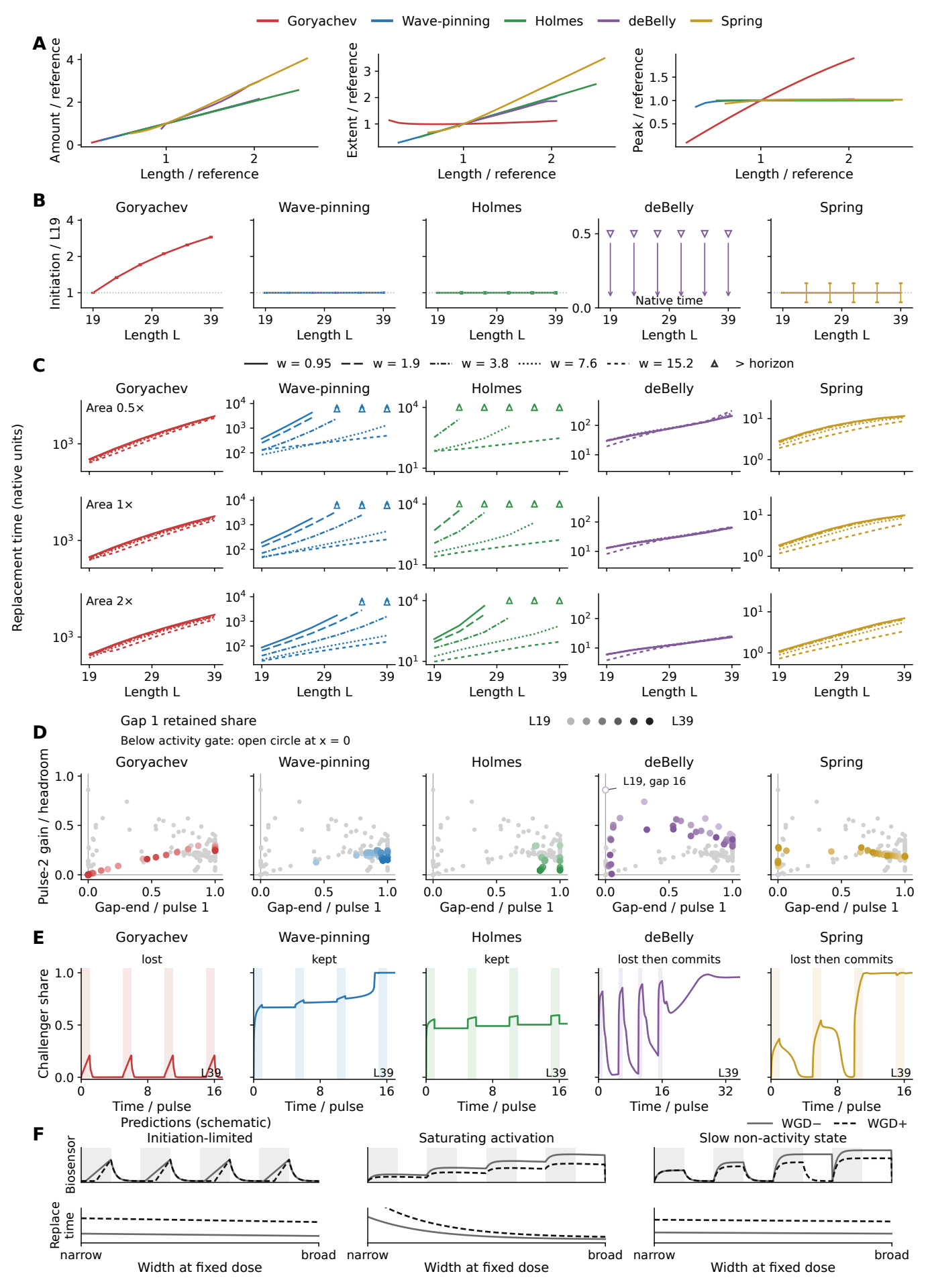
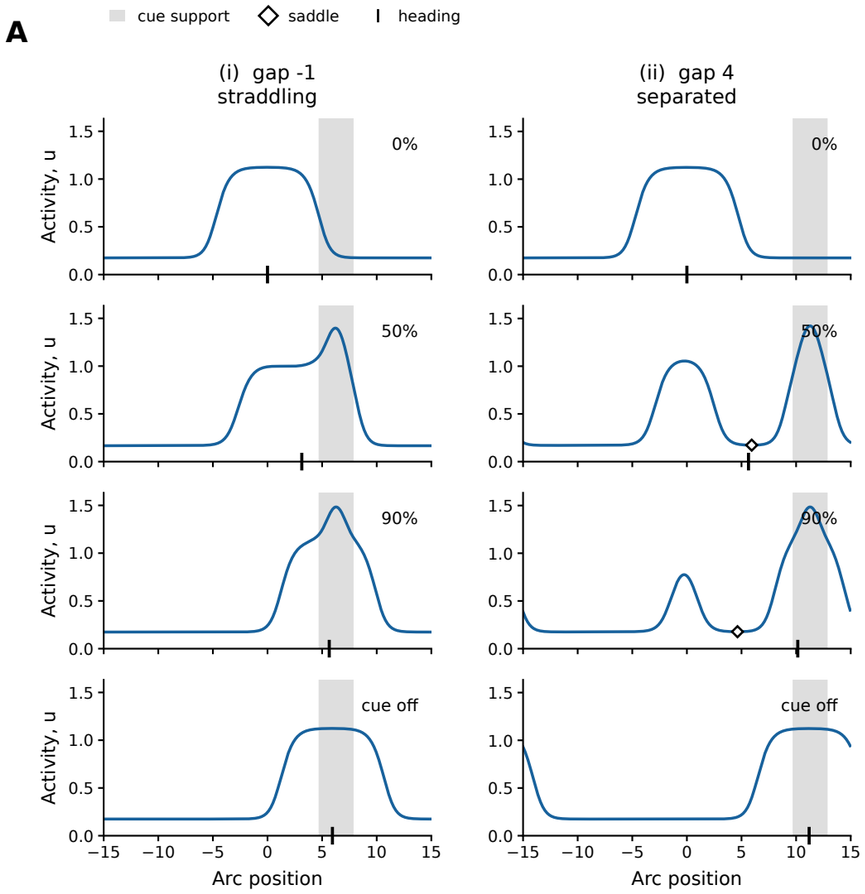
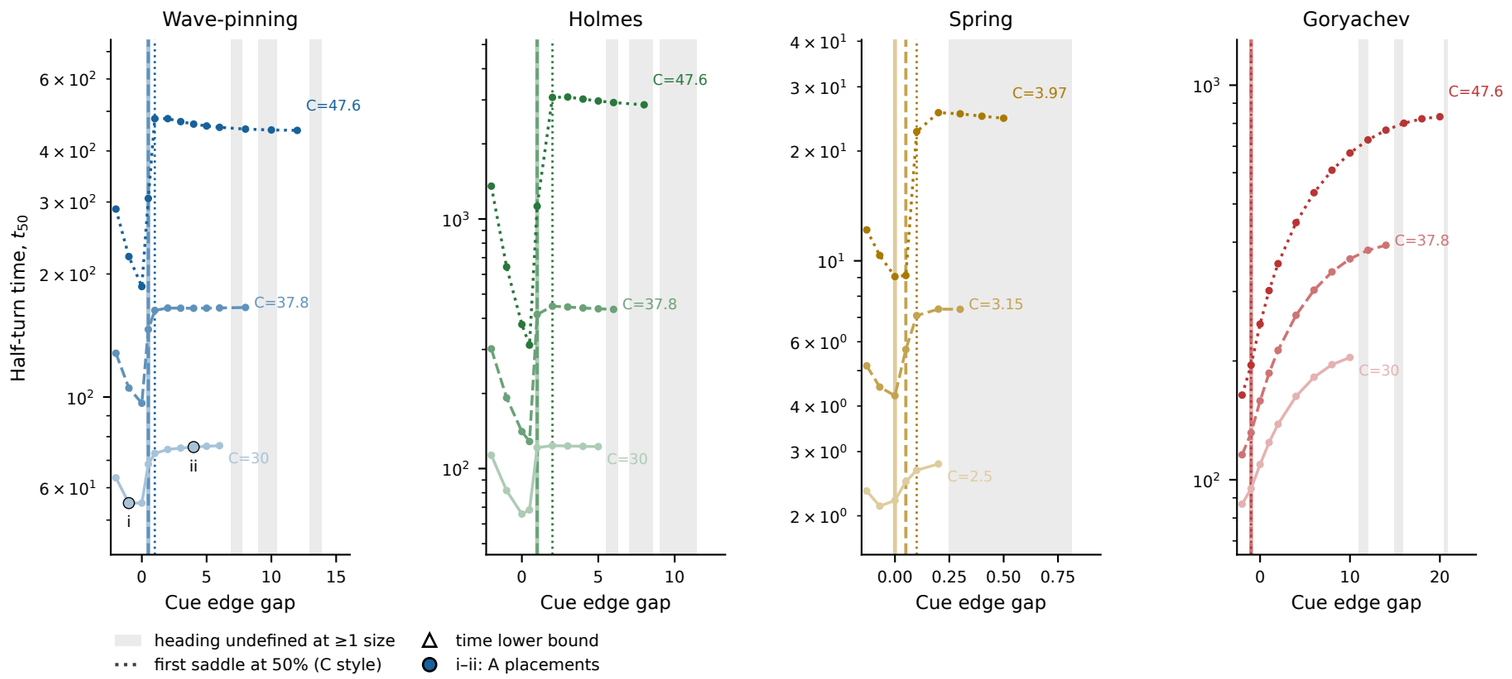
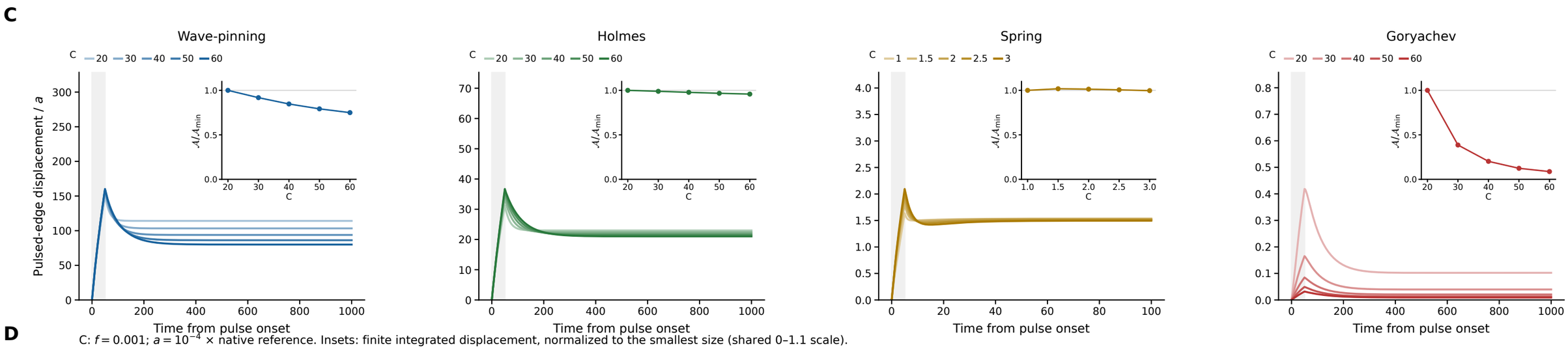

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose



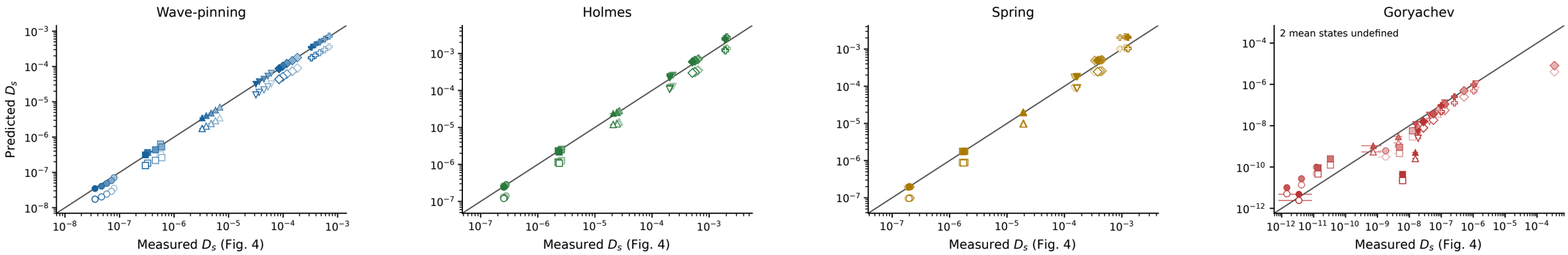


B: C = 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
C–D: chemical C = 20–60; Spring 1–3. These are not matched size grids.

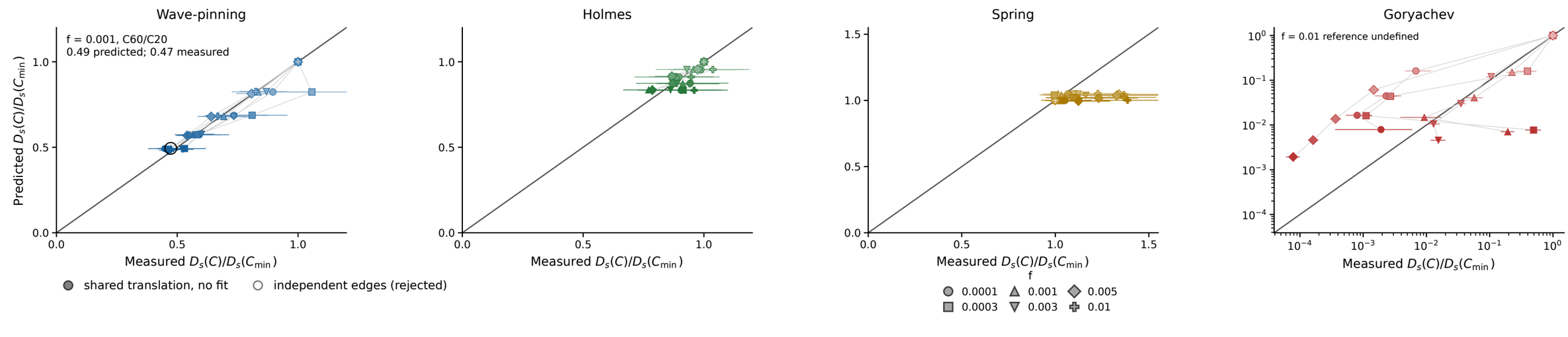


C: $f = 0.001$; $a = 10^{-4}$ × native reference. Insets: finite integrated displacement, normalized to the smallest size (shared 0–1.1 scale).

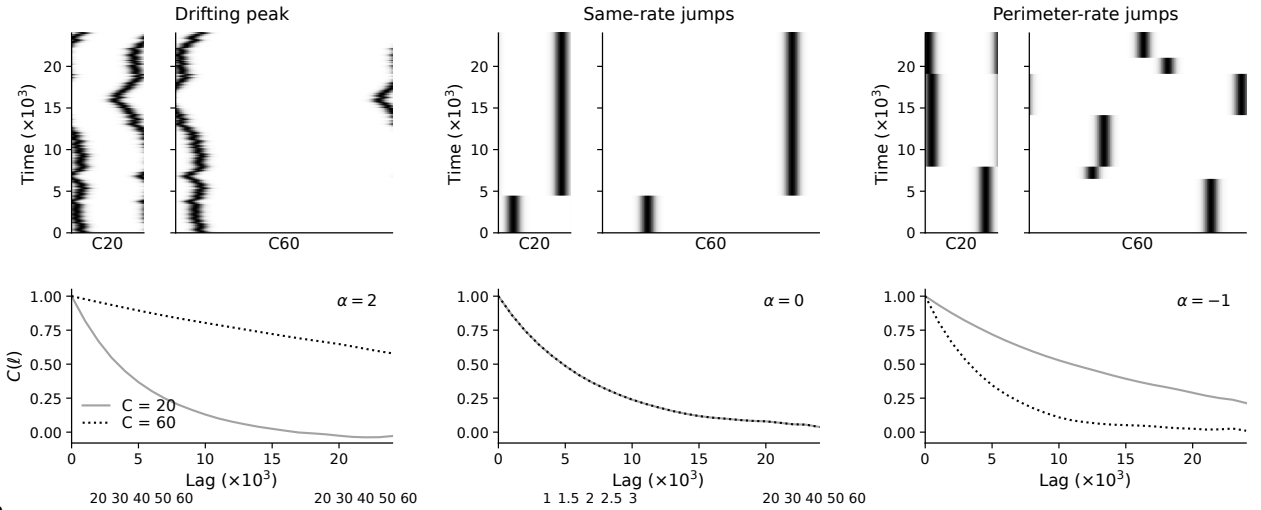
D



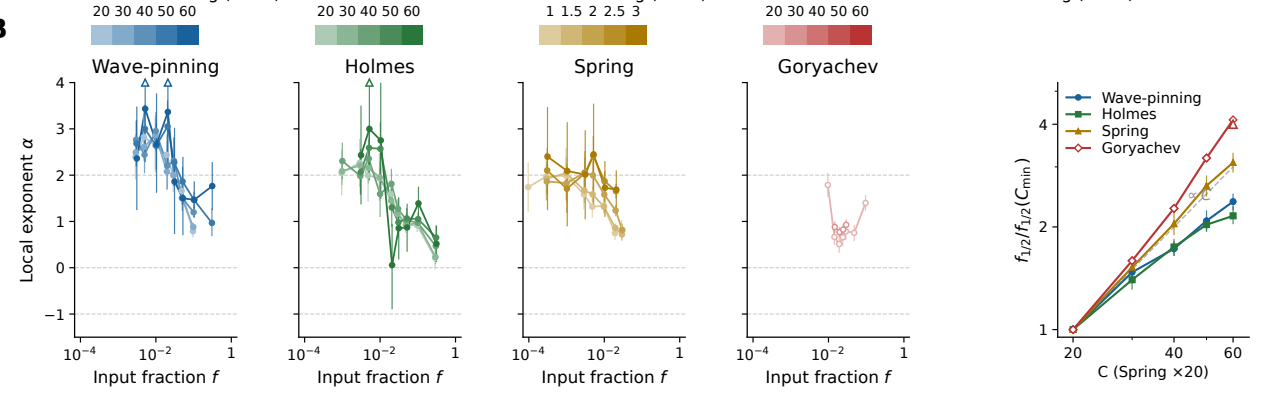
D: above, absolute diffusivity; below, size ratios at the same input. Lines: 1:1. Horizontal bars: whole-seed-bootstrap 95% CI.



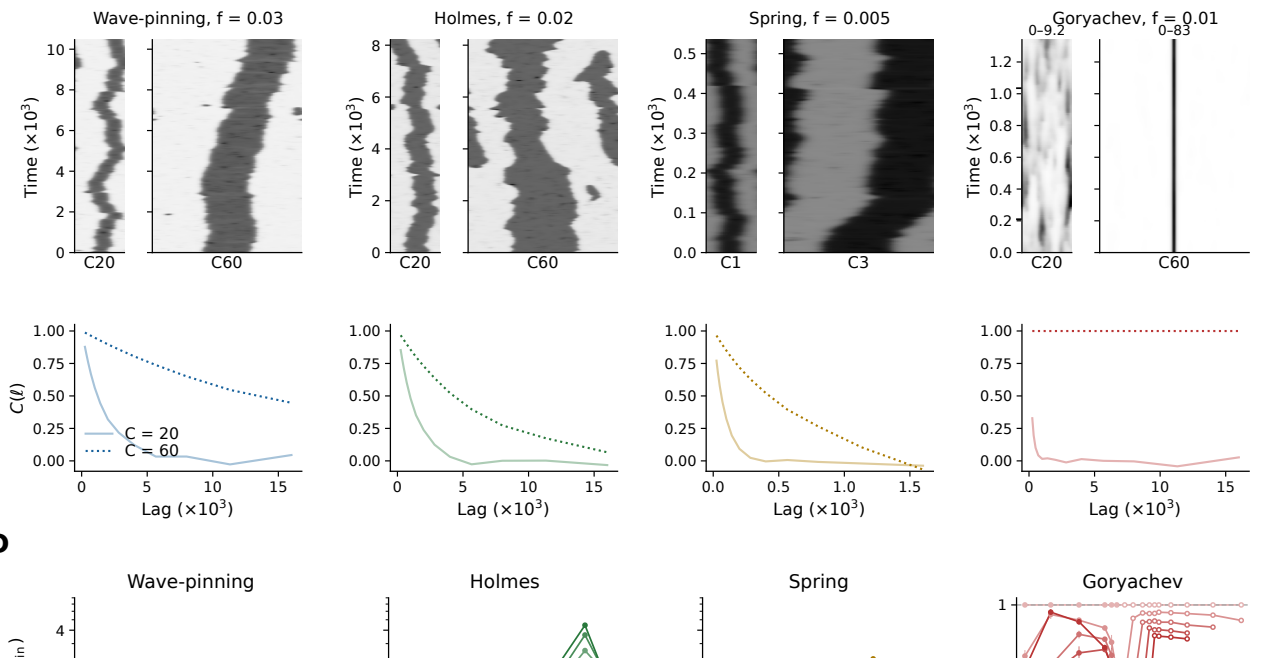
A


Drifting peak
Same-rate jumps
Perimeter-rate jumps
Time (×10³)
C20
C60
α = 2
α = 0
α = −1
C(ℓ)
C = 20
C = 60
Lag (×10³)

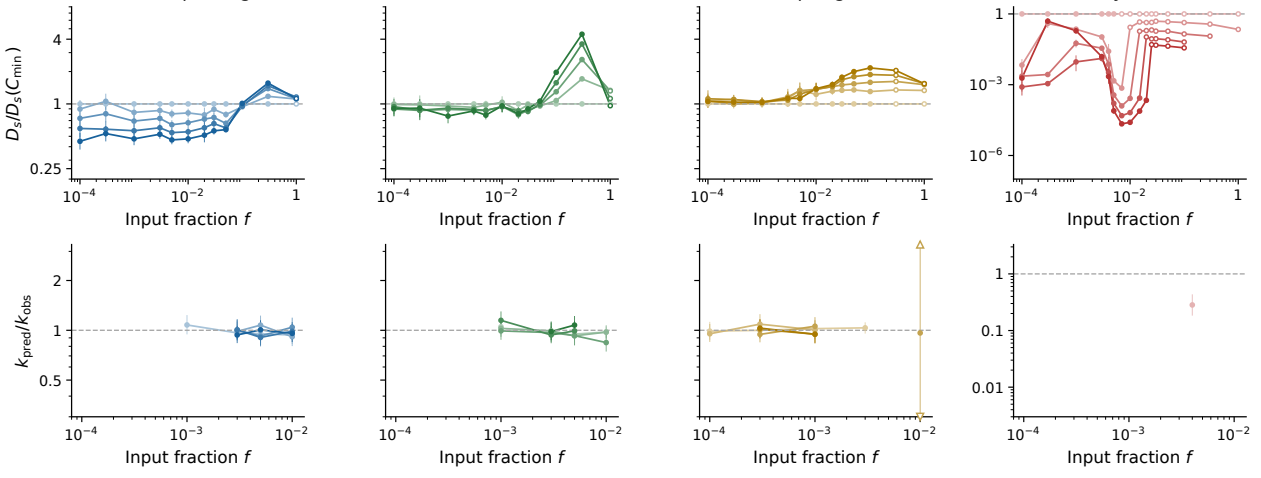
B


20 30 40 50 60
1 1.5 2 2.5 3
Wave-pinning
Holmes
Spring
Goryachev
Local exponent α
Input fraction f
Wave-pinning
Holmes
Spring
Goryachev
f1/2/f1/2(Cmin)
C (Spring ×20)

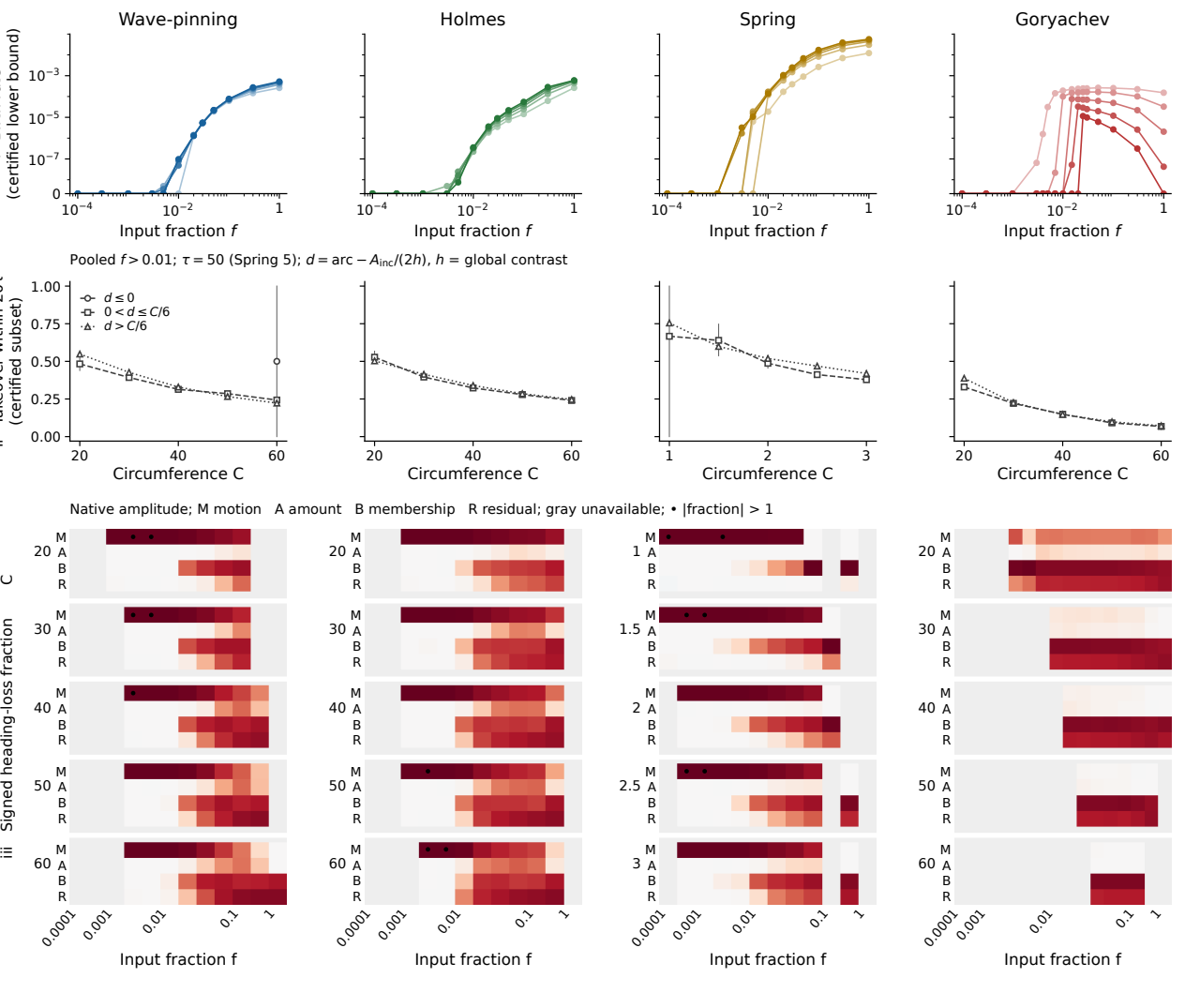
C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
0–9.2
0–83
Time (×10³)
C20
C60
C1
C3
C(ℓ)
C = 20
C = 60
Lag (×10³)

D


Wave-pinning
Holmes
Spring
Goryachev
Ds/Ds(Cmin)
kpred/kobs
Input fraction f

E


Wave-pinning
Holmes
Spring
Goryachev
i Birth rate (certified lower bound)
Input fraction f
Pooled f > 0.01; τ = 50 (Spring 5); d = arc − Ainc/(2h), h = global contrast
ii Takeover within 20τ (certified subset)
d ≤ 0
0 < d ≤ C/6
d > C/6
Circumference C
Native amplitude; M motion A amount B membership R residual; gray unavailable; • |fraction| > 1
iii Signed heading-loss fraction
C
M
A
B
R
Input fraction f